



CHINOOK MUSIC DATABASE

(A Data Analysis Report)

Abstract

This report is a data-driven analysis of the Chinook Music Database, conducted using MySQL Workbench. The key areas explored include customer behaviour, sales performance, artist revenue, and employee productivity. Using structured SQL queries involving joins, aggregations, window functions, and common table expressions (CTEs), meaningful insights were extracted and presented with clear interpretations to support strategic business decision-making. This report is aimed at providing actionable recommendations for Chinook's business operations.

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EXECUTIVE SUMMARY

The Chinook Music Database is a digital music store that sells tracks and albums to customers around the world. The database provided contained information on artists, albums, tracks, customers, employees, and sales transactions. An analysis was carried out on the store's data to provide answers to important business questions about what was selling well, who was buying more, and how the business was performing overall.

Using MySQL Workbench, a series of SQL queries were run to dig into the data and identify trends in the artists, albums, genres, customers, and sales tables. This was done to derive patterns and insights that could help the business make better decisions to drive growth.

The main objectives of the analysis were to:

- Identify the top performing artists, albums, and genres by revenue.
- Understand customer purchasing behaviour and spending patterns.
- Analyse monthly sales trends over time.
- Evaluate how tracks and genres contribute to overall revenue.

Some of the key insights gotten from the analysis include which artists and albums brought in the most money, which genres were most popular, and how customer spending varied across different months. These findings helped create a clearer picture of Chinook's business performance and highlighted areas where improvements and investments could be made to increase sales and productivity.

Q1 – Top 5 Artists by Total Revenue from Track Sales

Objective:

To identify which 5 artists generate the most revenue for Chinook through track sales.

Screenshot of Query:

```
3      -- Top 5 Artists In Terms of Sales:
4      SELECT ar.Name AS Artist_Name,
5             SUM(il.UnitPrice * il.Quantity) AS Total_Revenue
6      FROM artist ar
7      JOIN album al ON ar.ArtistId = al.ArtistId
8      JOIN track t ON al.AlbumId = t.AlbumId
9      JOIN invoiceline il ON t.TrackId = il.TrackId
10     GROUP BY ar.ArtistId, ar.Name
11     ORDER BY Total_Revenue DESC
12     LIMIT 5;
```

Screenshot of Result:

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	Artist_Name	Total_Revenue				
▶	Iron Maiden	138.60				
	U2	105.93				
	Metallica	90.09				
	Led Zeppelin	86.13				
	Lost	81.59				

Interpretation:

Iron Maiden led all artists with \$138.60 in total revenue, followed closely by U2 at \$105.93 and Metallica at \$90.09. The top 5 artists shown in the results are all classic rock and metal acts, suggesting that Chinook's customer base has a strong preference for this genre.

Since these artists are responsible for the highest track sales in the Chinook database, they should be prioritised in Chinook's marketing campaigns and promotional efforts as they have stood out as the strongest revenue drivers.

Q2 – Monthly Total Revenue & Running Total Across Months Per Artist

Objective:

To track each artist's monthly revenue over time and observe how their cumulative earnings for grow across months.

Screenshot of Query:

```
14  -- Artists & Their Monthly Total Revenue and Running Total:
15  •  SELECT ar.Name AS Artist_Name,
16         DATE_FORMAT (i.InvoiceDate, '%Y-%M') AS Month,
17         SUM(il.UnitPrice * il.Quantity) AS Monthly_Revenue,
18         SUM(SUM(il.UnitPrice * il.Quantity))
19         OVER (PARTITION BY ar.Name
20              ORDER BY DATE_FORMAT(i.InvoiceDate, '%Y-%M')) AS Running_Total
21  FROM artist ar
22  JOIN album al ON ar.ArtistId = al.ArtistId
23  JOIN track t ON al.AlbumId = t.AlbumId
24  JOIN invoiceline il ON t.TrackId = il.TrackId
25  JOIN invoice i ON il.InvoiceId = i.InvoiceId
26  GROUP BY ar.Name, DATE_FORMAT(i.InvoiceDate, '%Y-%M')
27  ORDER BY ar.Name, Month;
```

Screenshot of Result & Output:

Result Grid				
Filter Rows:		Exports:	Wrap Cell Contents: ☐	
Artist_Name	Month	Monthly_Revenue	Running_Total	
AC/DC	2021-January	5.94	5.94	
AC/DC	2022-April	3.96	9.90	
AC/DC	2023-July	2.97	12.87	
AC/DC	2024-November	2.97	15.84	
Academy of St. Martin in the Fields & Sir Neville ...	2024-October	0.99	0.99	
Academy of St. Martin in the Fields, John Birch, ...	2022-March	0.99	0.99	
Academy of St. Martin in the Fields, Sir Neville ...	2022-April	0.99	0.99	
Academy of St. Martin in the Fields, Sir Neville ...	2024-November	0.99	1.98	
Accept	2021-January	1.98	1.98	
Accept	2022-April	0.99	2.97	
Accept	2023-July	0.99	3.96	
Accept	2024-November	0.99	4.95	
Adrian Leaper & Doreen de Feis	2023-July	0.99	0.99	
Aerosmith	2021-January	3.96	3.96	
Aerosmith	2022-April	2.97	6.93	
Aerosmith	2023-July	1.98	8.91	
Aerosmith	2024-November	0.99	9.90	

Result 1

Output

Action Output

#	Time	Action	Message
1	21:26:12	use chinook	0 row(s) affected
2	21:26:30	SELECT ar.Name AS Artist_Name, DATE_FORMAT (i.InvoiceDate, '%Y-%M') AS Month, SUM...	666 row(s) returned

Interpretation:

The output returned 666 rows that showed the monthly revenue and the cumulative earnings for each artist across all recorded months.

AC/DC generated \$5.94 in January 2021, and this grew to a cumulative total of \$15.84 by November 2024. The running total revealed that while some artists generated consistent monthly revenue, others appeared only occasionally. This suggested that certain artists had loyal customers who kept buying their music, while others experienced occasional spikes in sales.

Q3 – Track with Highest Revenue Per Album

Objective:

To find the best performing track in each album and see how much it contributes to the album's overall revenue.

Screenshot of Query:

```
29  -- For Each Album, What Track Generated The Most Revenue & What Percentage of Total Album Revenue Did It Bring In?
30  •  SELECT al.Title AS Album_Name, t.Name AS Track_Name,
31         SUM(il.UnitPrice * il.Quantity) AS Track_Revenue,
32         SUM(SUM(il.UnitPrice * il.Quantity))
33         OVER (PARTITION BY al.AlbumId) AS Album_Revenue,
34         ROUND(SUM(il.UnitPrice * il.Quantity) /
35               SUM(SUM(il.UnitPrice * il.Quantity))
36               OVER (PARTITION BY al.AlbumId) * 100, 2)
37         AS Percentage
38  FROM album al
39  JOIN track t ON al.AlbumId = t.AlbumId
40  JOIN invoice il ON t.TrackId = il.TrackId
41  GROUP BY al.AlbumId, al.Title, t.TrackId, t.Name
42  ORDER BY al.Title, Track_Revenue DESC;
```

Screenshot of Result & Output:



The screenshot shows a database query result grid and an output log. The result grid displays the following data:

Album_Name	Track_Name	Track_Revenue	Album_Revenue	Percentage
...And Justice For All	Blackened	0.99	5.94	16.67
...And Justice For All	...And Justice For All	0.99	5.94	16.67
...And Justice For All	Eye Of The Beholder	0.99	5.94	16.67
...And Justice For All	Harvester Of Sorrow	0.99	5.94	16.67
...And Justice For All	The Frayed Ends Of Sanity	0.99	5.94	16.67
...And Justice For All	To Live Is To Die	0.99	5.94	16.67
20th Century Masters - The Millennium Collectio...	The Zoo	0.99	4.95	20.00
20th Century Masters - The Millennium Collectio...	Loving You Sunday Morning	0.99	4.95	20.00
20th Century Masters - The Millennium Collectio...	Still Loving You	0.99	4.95	20.00
20th Century Masters - The Millennium Collectio...	I Can't Explain	0.99	4.95	20.00
20th Century Masters - The Millennium Collectio...	Taste Me Please Ma	0.99	4.95	20.00

The output log shows the following actions:

#	Time	Action	Message
1	11:29:48	use chinook	0 row(s) affected
2	11:44:03	SELECT al.Title AS Album_Name, t.Name AS Track_Name, SUM(il.UnitPrice * il.Quantity) AS ...	1984 row(s) returned

Interpretation:

The results showed that in most albums, all tracks contributed equally to the total revenue. For example, in Metallica's "...And Justice For All", each of the 6 tracks contributed exactly

16.67% of the album's total revenue. This indicated that customers bought a variety of tracks rather than just one popular song.

Q4 – Artists with Declining Monthly Revenue

Objectives:

To identify artists whose sales dropped from one month to the next.

Screenshot of Query:

```
44      -- Artists Whose Monthly Revenue Dropped Compared To The Previous Month:
45  • ⊕ WITH Monthly_Revenue AS (
57  ⊖ Lagged AS (
58      SELECT
59          Artist_Name,
60          Month,
61          Revenue,
62          LAG(Revenue) OVER (PARTITION BY Artist_Name
63                              ORDER BY Month) AS Previous_Month_Revenue
64      FROM Monthly_Revenue
65  )
66  SELECT *
67  FROM Lagged
68  WHERE Revenue < Previous_Month_Revenue
69  ORDER BY Artist_Name, Month;
```

Screenshot of Result & Output:

Result Grid				
Filter Rows:		Export:		Wrap Cell Content:
	Artist_Name	Month	Revenue	Previous_Month_Revenue
▶	AC/DC	2022-April	3.96	5.94
	AC/DC	2023-July	2.97	3.96
	Accept	2022-April	0.99	1.98
	Aerosmith	2022-April	2.97	3.96
	Aerosmith	2023-July	1.98	2.97
	Aerosmith	2024-November	0.99	1.98
	Alice In Chains	2024-November	0.99	1.98
	Amy Winehouse	2024-November	0.99	10.89
	Antônio Carlos Jobim	2023-July	0.99	1.98
	Antônio Carlos Jobim	2024-November	1.98	8.91
	Apocalyptica	2023-July	0.99	1.98
	Audioslave	2022-March	1.98	2.97
	Audioslave	2023-June	0.99	2.97
	BackBeat	2023-July	0.99	1.98
	Battlestar Galactica	2023-April	5.97	7.96
	Battlestar Galactica	2024-August	1.99	5.97
	Battlestar Galactica ...	2024-October	1.99	17.91

Result 2 x

Output

Action Output

#	Time	Action	Message
✓ 1	12:47:05	WITH Monthly_Revenue AS (SELECT ar.Name AS Artist_Name, DATE_FORMAT(i.InvoiceD...	207 row(s) returned

Interpretation:

Several artists showed a consistent decline in monthly revenue over time. AC/DC dropped from \$5.94 in January 2021 to \$2.97 by November 2024.

Amy Winehouse experienced the sharpest single drop, falling from \$10.89 to just \$0.99 in one month. This suggested that customer interest was fading for these artists.

Chinook should consider promotions or discounts on these artists' tracks to bring sales back up.

Q5 – Top Selling Artist Per Genre

Objectives:

To find out which artist dominates sales in each music genre.

Screenshot of Query:

```
71      -- Top-Selling Artist For Each Genre Based On Total Revenue:
72  ● WITH Genre_Artist_Revenue AS (
73      SELECT
74          g.Name AS Genre,
75          ar.Name AS Artist_Name,
76          SUM(il.UnitPrice * il.Quantity) AS Total_Revenue
77      FROM genre g
78      JOIN track t ON g.GenreId = t.GenreId
79      JOIN invoiceline il ON t.TrackId = il.TrackId
80      JOIN album al ON t.AlbumId = al.AlbumId
81      JOIN artist ar ON al.ArtistId = ar.ArtistId
82      GROUP BY g.Name, ar.Name
83  ),
84  ○ Ranked AS (
85      SELECT
86          Genre,
87          Artist_Name,
88          Total_Revenue,
89          ○ RANK() OVER (PARTITION BY Genre
90                          ORDER BY Total_Revenue DESC) AS Rank_Num
91      FROM Genre_Artist_Revenue
92  )
93  SELECT
94      Genre,
95      Artist_Name,
96      Total_Revenue
97  FROM Ranked
98  WHERE Rank_Num = 1
99  ORDER BY Total_Revenue DESC;
```

Screenshot of Result & Output:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Genre	Artist_Name	Total_Revenue
▶	Metal	Metallica	90.09
	Rock	U2	90.09
	TV Shows	Lost	45.77
	Latin	Os Paralamas Do Sucesso	44.55
	Drama	Lost	35.82
	Sci Fi & Fantasy	Battlestar Galactica (Classic)	35.82
	Alternative & Punk	Titãs	33.66
	Blues	Eric Clapton	26.73
	Jazz	Spyro Gyra	19.80
	Comedy	The Office	17.91
	Pop	U2	15.84
	Reggae	Cidade Negra	15.84
	Bossa Nova	Toquinho & Vinícius	14.85
	Science Fiction	Battlestar Galactica	11.94
	Heavy Metal	Iron Maiden	11.88
	R&B/Soul	James Brown	11.88
	R&B/Soul	Amy Winehouse	11.88

Result 3

Output

Action Output

	#	Time	Action	Message
✓	1	12:47:05	WITH Monthly_Revenue AS (SELECT ar.Name AS Artist_Name, DATE_FORMAT(i.InvoiceD...	207 row(s) returned
✓	2	12:52:45	WITH Genre_Artist_Revenue AS (SELECT g.Name AS Genre, ar.Name AS Artist_Name,...	26 row(s) returned

Interpretation:

The results revealed a clear leader in every genre. Metallica topped the Metal genre with \$90.09 while U2 led Rock with the same amount. Lost and Battlestar Galactica appeared as top sellers in TV Shows, Drama and Sci Fi categories, showing that TV soundtracks were a strong revenue stream for Chinook. Eric Clapton led Blues and Spyro Gyra topped Jazz.

These findings suggest that Chinook can create genre-specific playlists and target promotions for each category.

CONCLUSION

This analysis of the Chinook Music Database revealed some really interesting patterns in how the business performed. Rock and Metal were clearly the strongest genres, with artists like Iron Maiden, U2 and Metallica consistently generating the highest revenue. TV show soundtracks also performed surprisingly well, which was something Chinook could have leaned into more.

Monthly sales trends showed that revenue was fairly stable over time, but some artists experienced declining sales. Acting early with promotions or recommendations could have helped maintain steady income.

Overall, the data painted a picture of a business with a loyal customer base and a strong catalogue. With the right focus on the top-performing artists and genres, Chinook has a great opportunity to grow its revenue even further.